

Candidate Number _____ Fraction _____

The **32nd** International Biology Olympiad National Team Selection Camp in **2021**

Practical Application of Cell and Plant Biotechnology **A**

Materials and experimental equipment:

Experimental	quantity
materials: (DNA1, DNA2) Blood tree DNA samples 1 and 2 (placed on ice)	1 tube each
PCR products A1, A2, B3, B4, C5, and C6 (placed on ice)	1 tube each
B. Experimental	quantity
Equipment: Microdispenser	1 branch each
(P2/P20), Micropipette tips (0.2µL~2µL),	1 box
Micropipette tips (2µL~200µL), 200µl	1 box
microcentrifuge tube	1
rack, 200µl	6 tubes
microcentrifuge tubes, Agarose gel (containing	1 piece
SYBR® Green),	1 piece
Parafilm, Oil-	1
based	1 box
pen, scissors, stapler	1
C. Drug solution:	quantity
(H2O) sterile water (placed on ice)	1 tube
(10X PCR) 10x PCR buffer (containing MgCl ₂) (keep on ice) (dNTPs)	1 tube
2 mM dNTPs (keep on ice)	1 tube
(Taq) Taq DNA polymerase (5 U/µl) (Keep on ice)	1 tube
(A primer) A. Primer atpB89 (2 µM) + rbcL188 (2 µM) (place on ice)	1 tube
(B primer) B. Primer trnL (2 µM) + trnF (2 µM) (place on ice)	1 tube
(C primer) C. Primer SSR F592 (2 µM) + R902 (2 µM) (Keep on ice) (dye)	1 tube
Tracking dye (Keep on ice) (oil) Mineral oil	1 tube
(Keep on ice) (100 bp DNA ladder) (Keep	1 tube
on ice)	1 tube
D. Shared instrument	quantity
benchtop microcentrifuge horizontal	5 units
electrophoresis tank	5 units
Thermal Cycler coating system	3 units
with UV lamp box	1 unit

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the question paper) consists of 5 pages and must be returned in its entirety when submitting the exam.
3. The total time for answering questions in this exam, including both Practice **A** and Practice **B**, is **90** minutes.
4. Please answer the questions on this paper. You may write your answers on the back, but please indicate the question number.

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

Part 1: Polymerase chain reaction amplifies specific **DNA** fragments from plants.

Wear latex gloves throughout the experiment.

Polymerase chain reaction (PCR) is a process that uses DNA polymerase to perform a reaction on a specific gene.

Massive amplification and synthesis in vitro can replicate a specific DNA fragment into ten billion to one thousand times its original size in a short period of time.

A billion times.

This practical experiment used the whole genome DNA of the dicotyledonous plant "Spatholobus suberectus" as material to perform intergenic analysis of 2 chloroplast gene fragments and 1...

PCR was performed on microsatellite loci, and the relationship between DNA amplification results and microsatellite loci was inferred from the PCR results.

Differences between samples.

Experimental steps:

I. Preparation and Operation of Polymerase Chain Reaction

1. Experiments were conducted on DNA samples 1 and 2 from "Pterocarya stenoptera", including two sets of chloroplast gene fragment primer pairs and one set of microsatellites.

(SSR1) locus primer pairs, a total of 6 tubes of PCR reaction solution were prepared. The experiment was performed on ice. After preparation...

Place it on ice and notify the teaching assistant to prepare for the next reaction.

2. Prepare the PCR reaction solution according to the table below : **(Please label each 200µl centrifuge tube with "Exam Number + A1, A2, B3, B4, C5, C6")**

PCR reaction solution	Exam number A1	Exam number A2	Exam number B3	Exam number B4	Exam number C5	Exam number C6
Sterile water (1.5 ml microcentrifuge tube) 10.5 µl	10.5 µl	10.5 µl	10.5 µl			
10-fold concentration PCR buffer 2 µl	2 µl	2 µl	2 µl		2 µl	
2 mM dNTP	2 µl	2 µl	2 µl		µl	
Taq DNA polymerase (5Units/ul)	0.5 µl	0.5 µl	0.5 µl	0.5 µl	0.5 µl	0.5 µl
A. Primer atpB89 (2 µM)+ rbcL188 (2 µM) Chloroplast gene fragments	4 µl	4 µl				
B. Primer trnL (2 µM)+ trnF (2 µM) Chloroplast gene fragments			4 µl	4 µl		
C. Primer SSR F652 (2 µM)+ SSR R902 (2 µM) microsatellite loci					4 µl	4 µl
Bloodwood DNA Sample 1	1 µl		1 µl		1 µl	
Bloodwood DNA Sample 2		1 µl	1 µl	1 µl		
Mineral oil	Appropriate amount cover	Appropriate amount cover	Appropriate amount cover	Appropriate amount cover	Appropriate amount cover	Appropriate amount cover
Total volume 20 µl	20 µl	20 µl	20 µl	20 µl	20 µl	20 µl

After adding the PCR materials, briefly centrifuge the tubes and then transfer them to the public instrument area, where the teaching assistant will place them in a circulating temperature-controlled environment.

The instrument was in use. After the reaction was completed, the teaching assistant retrieved it.

3. Place the reaction in a thermal cyclone controller.

94℃ 16s 94℃ 54℃ 72℃ 72℃
2 minutes → 15 seconds → 15 seconds → 30 seconds → 2 minutes →

~~~~~  
30 cycles

No need to wait for PCR results, please perform the following experiments immediately!!

II. Agarose gel electrophoresis analysis

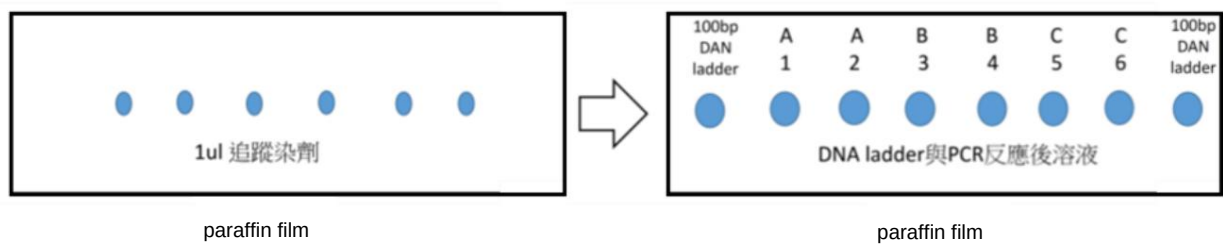
1. Apply 1μl of tracking dye to the paraffin film each time, for a total of 6 times, distributing it evenly across the paraffin film.

superior.

2. Take 5μl of each of the separately prepared (A1, A2, B3, B4, C5, C6) PCR products from the teaching assistant and mix them with 1μl of tracking dye.

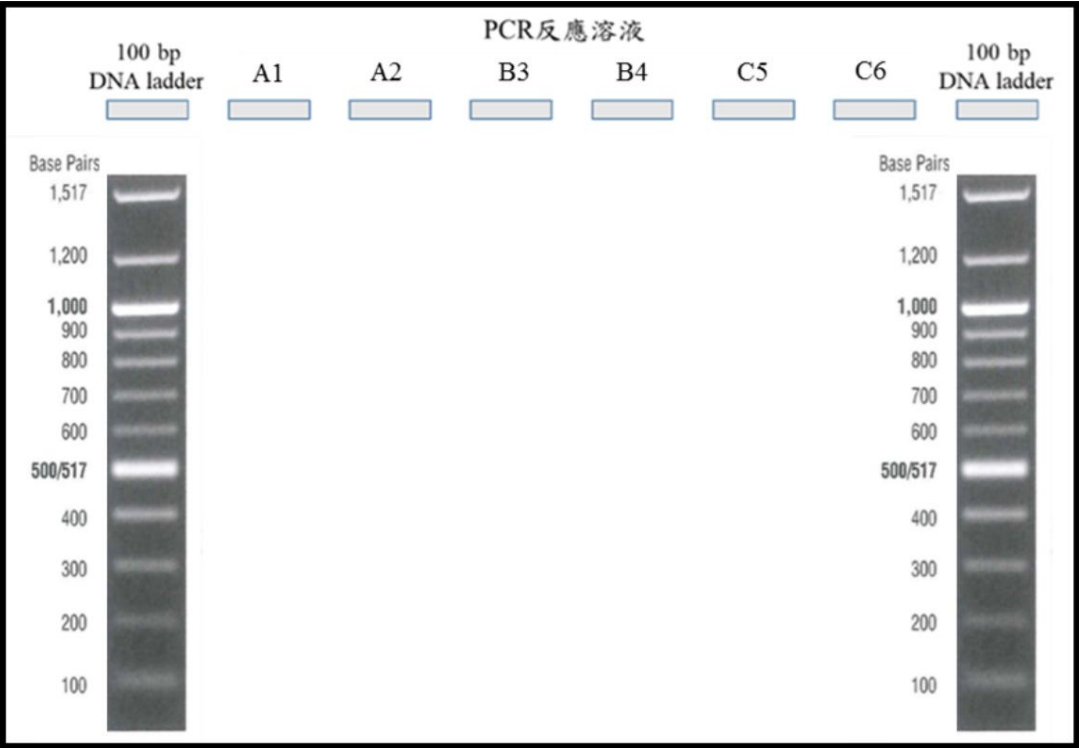
(tracking dye)→

3. Take 2 drops of 3μl 100 bp DNA ladder (containing tracking dye) and place them on the designated position on the paraffin membrane.



4. Take 3μl of 100 bp DNA ladder and 5μl of the PCR reaction solution after mixing in the above steps, from left to right.

Slowly inject the sample into the groove of the electrophoresis gel (as shown in the figure below).



5. After completion, take the electrophoresis film to the common area, where the teaching assistant will place it into the electrophoresis tank for the gel running process and collect the timing sheet.

After 25 minutes, retrieve the electrophoresis gel from the teaching assistant.

Gel running conditions: 170V voltage causes DNA to move towards the positive electrode until the tracking dye is 1/3 of the way from the bottom.

6. After the gel is run, please take the electrophoresis film to the gel photo system in the public area and have the teaching assistant take a picture of it and print out the results.

Results and

Discussion: 1. Please attach the electrophoresis film images. Based on the observed results, draw a simplified 100 bp DNA ladder and agarose gel, and label the name of each reaction sample and the estimated size of the DNA fragment. Clearly label the corresponding electronic image file names as well. (20%)

2. The numerical designations of the forward SSRs F652 and R902 in the microsatellite locus-specific primer set represent the start position of the primer sequence at the standard sample locus, which can be used to estimate the probable fragments of this PCR product.

2.1. Estimate

the possible fragment size of the PCR product from the standard sample locus? (5%) 2.2.

Based on the observed DNA fragment band size from the microsatellite locus in Experiment 2, infer the size of this fragment.

Possible reasons for the formation of this DNA band in the two samples. (15%)

3. Based on the following original reagent solution concentrations, estimate the concentrations of each component in the reaction solution: (10%)

| PCR reaction solution        | Exam number<br>A1 | Exam number<br>A2 | Exam number<br>B3 | Exam number<br>B4 | Exam number<br>C5 | Exam number<br>C6 |
|------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10x PCR buffer               |                   |                   |                   |                   |                   |                   |
| 2 mM dNTP                    |                   |                   |                   |                   |                   |                   |
| Taq DNA polymerase           |                   |                   |                   |                   |                   |                   |
| A. Primer atpB89 / rbcL188   |                   |                   |                   |                   |                   |                   |
| B. Primer trnL / trnF        |                   |                   |                   |                   |                   |                   |
| C. Primer SSRF652 / SSR R902 |                   |                   |                   |                   |                   |                   |

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## Cell and plant biotechnology

Practice **B**

Materials and experimental equipment:

| A. Experimental plant                                                                  |                    |                                        | quantity     |
|----------------------------------------------------------------------------------------|--------------------|----------------------------------------|--------------|
| materials: Begonia plants, chive plants                                                |                    |                                        | 1 share each |
| Broad bean root tips were fixed in a fixative solution and then stored in 70% alcohol. |                    |                                        | 1 pack       |
| B. Experimental Equipment:                                                             | Quantity 11.       | Filter paper (1/4 size) 1.             | 8 pieces     |
| Duplex microscope + electron eyepiece + laptop (1 set) 12.                             |                    | Waste liquid cup                       | 1            |
| 2. Single-sided blade                                                                  | 2 pieces,          | 13. Trash can                          | 1            |
| 3. Double-sided blade                                                                  | 3 sheets of        | paper towels and toilet paper.         | Group 1      |
| 4. Petri dish                                                                          | 2 x 15.            | Washing bottles                        | 1 bottle     |
| 5. Tweezers                                                                            | 2 vials of         | C drug solution:                       | quantity     |
| 6. Plastic dropper                                                                     | 5 tubes of         | 1. Mixed dye (Alcian blue, Safranin O) | 1 bottle     |
| 7. Cover/slide                                                                         | Group 7,           | 2.1 mol L <sup>-1</sup> HCl            | 1 tube       |
| 8. Scissors                                                                            | 1 tube of          | 3. Lactic acid lichen red dye          | 1 bottle     |
| 9. Dissecting needle                                                                   | 1 D-type           | shared instrument                      | quantity     |
| 10. Oil-based ballpoint pen                                                            | 1 dry bath at 1.60 | ÿ                                      | 1 unit       |

Please note:

1. The total time allotted for this exam, including both Practice A and Practice B, is 90 minutes.
2. The materials and equipment on the table will not be replenished once they are used up.
3. This exam paper (including the cover and the exam paper) has a total of 7 pages and must be returned in its entirety when submitting the exam.
4. Please answer the questions on this paper. You may write your answers on the back, but please indicate the question number.

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

## Part One utilizes freehand sectioning and staining to observe plant tissues.

The blades used are very sharp. Please be careful when performing hand slicing. If you are injured, please immediately notify the teacher or teaching assistant present.

The freehand sectioning method involves using a single-edged razor blade to cut a thin section of the material to be observed, and then placing the section in water.

In a petri dish, use tweezers to select an appropriate thin slice, place it on a glass slide, and observe it using an optical microscope.

To increase tissue contrast, staining can be applied. The advantage of this method is its simplicity and convenience, but the disadvantage is the quality.

Significant ambiguity (e.g., uneven thickness of the slide material) generally necessitates a slide thickness of around 25  $\mu\text{m}$ . This practical exercise involved freehand sectioning of leaves and

stems of begonias (dicotyledonous plants) and chives (monocotyledonous plants).

Sections were prepared and tissue staining was used to understand the differences in tissues among different types of plants.

Experimental steps:

1. Harvest the leaves and stems of begonias and chives.
2. Before sectioning, place a small amount of water in the petri dish, and prepare tweezers, a dropper, and a single-edged scalpel, etc.

The materials observed.

3. Place the material on a glass slide and cut a thin section of appropriate thickness using a single-edged blade (Figure 1). If it is not possible to obtain an ideal thin section directly with a single-edged blade, three double-edged blades placed side by side can be used, with the middle blade...

Slightly lift and keep both blades parallel, using the gap between the blades to obtain the subsequent...

Thin slices are required for this step (Figure 2). Slicing should be done quickly, cutting the material in one go (the entire process...).

When cutting, the material and blade should be moistened with clean water to make them smooth; otherwise, the material is easily damaged.



Figure 1, bare-handed



Figure 2, 3 blades

4. After cutting many thin slices by repeating this process, gently transfer these slices into a petri dish filled with water using tweezers.

Prepare for later use (when slicing by hand, the most important thing is to cut off a small, flat, and thin piece of tissue).

5. Select three of the best cross-sections from the petri dish, place the sections in another empty petri dish, and add one...

Apply the mixed dye for ten seconds.

6. Use distilled water from the washing bottle to wash away any excess mixed dye.
7. Use tweezers to select a satisfactory section and place it on a glass slide. Add a drop of water, cover with a coverslip, and then use filter paper.

Wipe away any excess moisture and observe under a microscope.

8. Take photos using an electronic eyepiece for scoring purposes.

(1) Save to desktop: Exam Number Folder. (2) File name: Exam Number + (Crabapple Leaf, Crabapple Stem, Leek Leaf, Leek Stem) each represents one of them. (3) For multiple photos of the same part, please add numbers, for example: Exam Number + Crabapple Leaf 1, Exam Number + Crabapple Leaf 2.

9. Take another material and repeat the above steps.

Results and Discussion:

1. Based on the observations, draw simple sections of leaf and stem tissues of begonia and chives, and label the tissue names and...

The file name of the photo corresponding to its slice.

A. Begonia - Leaf (7%) (Drawing 5 points, Electronic Eyepiece 2 points) Please draw a simple

| section of Begonia - Leaf - Tissue Section | Electronic eyepiece scoring |
|--------------------------------------------|-----------------------------|
|                                            | Filled out by the teacher   |

B. Begonia - Stem (7%) (Drawing 5 points, Electronic eyepiece 2 points)

| Please draw a simple diagram of a begonia stem tissue section. | Electronic eyepiece scoring |
|----------------------------------------------------------------|-----------------------------|
|                                                                | Filled out by the teacher   |



C. Leek leaves (7%) (Drawing 5 points, electronic eyepiece 2 points)

|                                                              |                             |
|--------------------------------------------------------------|-----------------------------|
| Please draw a simple diagram of a chive leaf tissue section. | Electronic eyepiece scoring |
|                                                              | Filled out by the teacher   |

D. Leek stem (7%) (Drawing 5 points, electronic eyepiece 2 points)

|                                                     |                             |
|-----------------------------------------------------|-----------------------------|
| Please draw a simple tissue section of a leek stem. | Electronic eyepiece scoring |
|                                                     | Filled out by the teacher   |

2. Based on the observations, explain the differences between the leaf and stem tissues of dicotyledonous and monocotyledonous plants.

(10%)

## Part Two: Observation of cell division in broad bean root tips.

Mitosis is the primary mode of reproduction in eukaryotic cells. In eukaryotic cells, chromosomes in the nucleus divide equally.

The cells are divided into two daughter nuclei. This is called mitosis because a spindle-shaped structure appears during cell division.

A series of changes, including the formation of mitosis and chromosomes, occur before the cell truly divides into two. Mitosis is the process by which a cell divides into two.

Nuclear division is usually followed by cytokinesis, which divides the cytoplasm, organelles, cell membrane, and other cellular structures equally.

Distribution to daughter cells. This is a continuous, complex, and dynamic process. For ease of description, it is based on morphological changes.

The process of cleavage is divided into interphase (the stage of non-division) and cleavage (the stage of ongoing division) according to the sequence of these processes.

This practical experiment aims to identify the root tips of rapidly growing broad beans in different locations by combining tissue staining.

Root tip cells of broad bean during mitosis.

Experimental steps:

1. Use a dropper to remove the alcohol from the fixed broad bean sample with the reference number, and replace it with 1 mL of HCl.

Place in a 60°C dry bath for 5 minutes (to soften cells), and finally store in a wash bottle with distilled water.

2. Use tweezers to remove the root tip and place it on a clean glass slide. Add one drop of lactic acid lichen red stain to the root tip.

superior.

3. Gently tap the material vertically up and down with the thicker end of the dissecting needle (raise the thicker end to 1 cm above the slide, then...)

Then let it fall freely (repeated 20 times) to produce suspended single cells (Figure 3).

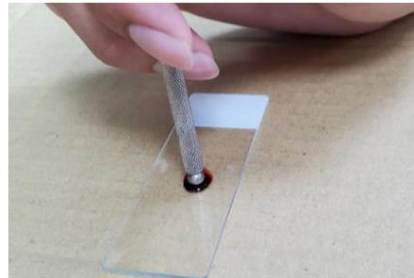
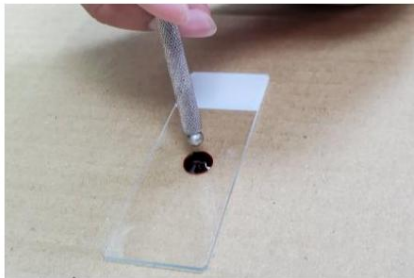


Figure 3. Gently tapping the material with the thicker end of a dissecting needle.

4. Use tweezers to remove any visible solid particles and residue.

5. Add 1 drop of lactic acid lichen red stain, cover with a coverslip, and then cover the slide with a paper towel. Hold the coverslip vertically with your thumb.

Apply pressure to flatten the cells.

6. Observe the different stages of mitosis under a microscope and take pictures using an electronic eyepiece to record the data.

(Archived: Examination Number Folder. File Name: Examination Number + Root Tip 1, Root Tip 2, etc.) The longer

the staining time, the darker the staining, but if the time is too long, the chromosomes and background become difficult to distinguish; if the mammary gland...

If the concentration of acid lichen red dye is insufficient, place tissue paper on the edge of the coverslip to absorb the liquid, while simultaneously applying it to another...

Dye is dripped onto the surface to repeat the staining process.

## Results and Discussion

1. Briefly sketch the different stages of mitosis observed in broad bean root tip cells, and label the stages of mitosis on the diagram.

Phases and their corresponding photo file names. (12%) (Drawing 7 points, Electronic Eyepiece 5 points) (Archived: Exam

Number Folder. File name: Exam Number + Root Apex 1, Root Apex 2)

| Please briefly draw a diagram of the different mitotic stages observed in broad bean root tip cells. | Electronic eyepiece scoring |
|------------------------------------------------------------------------------------------------------|-----------------------------|
|                                                                                                      | Filled out by the teacher   |